13.1 Reading the Headlines

WORKSHEET

The question

STEP I 1992, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q14

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If deaths occur independently and at random at a steady rate, how much should the yearly total vary? If the average is 1080 a year, what is a typical deviation from year to year?

STAGE 1 "Six times safer to fly than to walk"

Covers: Part (i)

Ask yourself The ratio $1080/180 = 6$ compares total deaths. Is that the same as comparing risks? What would a fair measure of risk need to take into account?

STAGE 2 "Our skies are safer: only 125 killed in 1990"

Covers: Part (ii)

Ask yourself Are air deaths independent rare events, or do they come in clusters? How much would you expect the yearly total to vary? Can one year show a trend?

STAGE 3 "Road carnage increasing: 7 killed on Tuesday"

Covers: Part (iii)

Pause and try How many pedestrians are killed per day on average? If daily deaths are Poisson with that mean, how many days a year would you expect 7 or more? And can one day show that something is increasing?

STAGE 4 "Alarming rise: 1350 pedestrians killed in 1990"

Covers: Part (iv)

Ask yourself Under a Poisson model with mean 1080, how many standard deviations above the mean is 1350? If it is not chance, what else could explain it? Think about what an average over 16 years can hide.

What's really going on

Signal and noise

Every statistical question is, at bottom, a question of separating *signal* (a real effect or trend) from *noise* (random variation). The Poisson model gives a yardstick for the noise in counts of independent events: a standard deviation of $\sqrt{\mu}$. Rare events have large *relative* variation: a count of 9 varies by about ± 3, a third of its size, while a count of 10 000 varies by only about 1%. That is why small numbers (deaths on one day, crashes in one year) are so often over-interpreted, and why cancer "clusters" in small towns usually turn out to be chance.

Regression to the mean

There is a further trap. Road-safety cameras are often installed at sites that have just had an unusually bad year. The following year, accidents fall, and the cameras get the credit. But an unusually bad year is likely to be followed by a more ordinary one anyway: this is *regression to the mean*, first described by Francis Galton in the 1880s. Separating the effect of a camera from regression to the mean requires comparison with similar sites that did not get cameras, the idea behind controlled experiments.

Florence Nightingale's statistics

Florence Nightingale, famous as a nurse in the Crimean War, was also a pioneering statistician. Her "rose diagrams" of 1858 showed that far more soldiers were dying of preventable disease than of wounds, and persuaded the government to reform army sanitation. She insisted on rates (deaths per thousand soldiers per year) rather than raw totals, and on comparing like with like, the very points this question tests. She was the first woman elected a Fellow of the Royal Statistical Society.

Try this next

1. **Poisson tails.** Check that $\mathrm{P}(X \geq 7) \approx 0.031$ for X Poisson with mean 2.96. How many days a year would you expect 9 or more pedestrian deaths?

2. **Clustered deaths.** Suppose air crashes occur as a Poisson process at 1.2 per year, and each kills 150 people. Find the mean and standard deviation of yearly air deaths. Is a year with 125 deaths or fewer unusual? Could a year have exactly 125 deaths?

3. **A significant rise?** If the 1974–1989 figures had been steady at 1080 ± 33, find (using a normal approximation) the probability of 1350 or more in a single year.

4. **Write a better headline.** For each of the four headlines, write a headline that the data actually support.

Further reading

- **David Spiegelhalter, *The Art of Statistics*** (Pelican). How to learn from data, with many examples from the news.
- **Tim Harford, *How to Make the World Add Up*** (Bridge Street). Ten rules for thinking about numbers in the media.
- **Darrell Huff, *How to Lie with Statistics*** (Penguin). A short classic from 1954, still relevant.

The full guided solution is in Section 13.1:
https://beyondstep.keithmcnulty.org/ch13/headlines.html